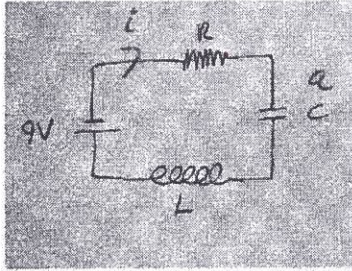


Wed 9

Name

Write down ODE for the circuit.



$$i = q'$$

$$V_R = iR \quad V_C = q/C \quad V_L = L i'$$

~~$$q + Rq' + q/C + Lq'' = 0$$~~

1. Write down and solve the ODE for the circuit with the inductor replaced by a wire.

$$q + Rq' + q/C = 0$$

$$q' + \frac{1}{RC} q = -q/R$$

lin 1st order
 $v(t) = e^{+t/RC}$

$$(e^{t/RC} q)' = -q/R e^{t/RC}$$

$$e^{t/RC} q = -q/R e^{t/RC} \cdot RC + \text{const}$$

$$q = -q/R \cdot RC + \text{const} e^{-t/RC}$$

$$q = -qC + \text{const} e^{-t/RC}$$

■ Solve $y' = 2y + e^t$ $y(0) = 1$

$$e^0(1) = -e^0 + C$$

$$C = 2$$

I.C.

Lin

$$y' - 2y = e^t \quad v = e^{-2t}$$

$$(e^{-2t}y)' = e^{-t}$$

$$e^{-2t}y = -e^{-t} + C$$

Gen
sol~~General solution~~2. Solve $y' + 2ty = te^{-t^2}$

Lin

$$v(t) = e^{\int 2t dt} = e^{t^2}$$

$$e^{t^2}y + 2te^{t^2}y = te^{-t^2}e^{t^2} = t$$

$$(e^{t^2}y)' = t$$

$$e^{t^2}y = \frac{t^2}{2} + C$$

$$y = \frac{t^2}{2}e^{-t^2} + Ce^{-t^2}$$